

AQbD Analytical Method Development Report

Analytical Quality by Design (AQbD) / Design of Experiments (DOE) Studio V1

Project Code:	Paracetamol_Dev	Date / Time:	2026-10-02 02:06:56
Project Name:	HPLC Paracetamol Method Development	Software:	AQbD Studio V1 (ICH-Aligned)

1. Executive Summary

This report documents the systematic, science- and risk-based development of the analytical procedure following ICH Q14 guidelines. The workflow progresses from defining the Analytical Target Profile (ATP) and initial Failure Mode and Effects Analysis (FMEA) risk assessment, to multivariate Design of Experiments (DOE), rigorous ANOVA and diagnostic model verification, multi-response desirability optimization, design space boundaries determination, and experimental confirmation.

2. Analytical Target Profile (ATP)

Parameter	Criterion	Unit	Acceptance Limits / Target
Retention Time	RANGE	min	[4.0 to 6.0]
Assay	TARGET	%	= 100.0
Tailing Factor	MAXIMUM	-	<= 2.0
Theoretical Plates	MINIMUM	-	>= 2000.0

3. Risk Assessment (FMEA)

Parameter	Severity (S)	Occurrence (O)	Detectability (D)	RPN	Priority
Mobile Phase Composition	8	6	5	240	Critical

4. Experimental Factors (DOE Parameters)

Code	Factor Name	Unit	Low (-1)	High (+1)	Role
A	Mobile Phase Composition	%	60.0	70.0	CRITICAL
B	Flow Rate	mL/min	0.8	1.2	CRITICAL
C	Column Temperature	°C	25.0	35.0	CRITICAL

5. Experimental Responses & Goals

Code	Response Name	Unit	Goal	Limits / Target	Weight
Y1	Resolution	-	MAXIMIZE	-	5
Y2	Tailing Factor	-	MINIMIZE	-	4
Y3	Retention Time	min	TARGET	= 5.0	4
Y4	Theoretical Plates	plates	MAXIMIZE	-	3

6. Design of Experiments (DOE) Execution

Design Type: **FULL_FACTORIAL** | Runs: **11** | Random Seed: **42**
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Design Type: **FULL_FACTORIAL** | Runs: **11** | Random Seed: **43**

7. Statistical Modeling & Optimization (Active Models)

Response Model (ID 5): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 10): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 11): **INTERACTIONS** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 12): **QUADRATIC** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 13): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 14): **INTERACTIONS** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 15): **QUADRATIC** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 16): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 17): **INTERACTIONS** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 18): **QUADRATIC** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 19): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 20): **INTERACTIONS** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 21): **QUADRATIC** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 24): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 25): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 26): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 27): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**
Response Model (ID 28): **FIRST_ORDER** | R²: **N/A** | Adj R²: **N/A** | Pred R²: **N/A**

Multi-Response Optimal Solution (Desirability):

Overall Desirability: **0.5000**
Factor Settings: A = 64.420, B = 0.884, C = 30.795
Overall Desirability: **0.5000**
Factor Settings: A = 66.343, B = 0.888, C = 34.364
Overall Desirability: **0.5000**
Factor Settings: A = 62.078, B = 1.072, C = 30.283
Overall Desirability: **0.5000**
Factor Settings: A = 63.268, B = 1.148, C = 28.182
Overall Desirability: **0.5000**
Factor Settings: A = 62.149, B = 1.168, C = 31.562
Overall Desirability: **0.5000**
Factor Settings: A = 62.517, B = 1.098, C = 32.057
Overall Desirability: **0.0000**
Factor Settings: A = 67.023, B = 1.037, C = 26.737
Overall Desirability: **0.0000**
Factor Settings: A = 67.023, B = 1.037, C = 26.737
Overall Desirability: **0.0000**
Factor Settings: A = 67.023, B = 1.037, C = 26.737
Overall Desirability: **0.0000**
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Overall Desirability: **0.0000**
Factor Settings: A = 67.023, B = 1.037, C = 26.737
Overall Desirability: **0.6253**
Factor Settings: A = 66.271, B = 0.800, C = 35.000
Overall Desirability: **0.0000**
Factor Settings: A = 67.023, B = 1.037, C = 26.737
Overall Desirability: **0.0000**
Factor Settings: A = 67.023, B = 1.037, C = 26.737
Overall Desirability: **0.0000**
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8. Confirmation Validation

Confirmation Run ID: 3 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
- Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
- Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
- Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000

Confirmation Run ID: 4 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
- Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
- Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
- Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000

Confirmation Run ID: 5 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
- Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
- Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
- Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000

Confirmation Run ID: 6 — ■ Software Simulation Check

- Y1: Predicted = 2.599 | Simulated = 2.599 | Diff = 0.000
- Y2: Predicted = 1.101 | Simulated = 1.101 | Diff = 0.000
- Y3: Predicted = 5.324 | Simulated = 5.324 | Diff = 0.000
- Y4: Predicted = 2698.378 | Simulated = 2698.378 | Diff = 0.000

9. Conclusion

The analytical method was developed and evaluated using systematic AQbD principles. The mathematical models, design space boundaries, and confirmation verification prove that the procedure operates robustly within its acceptable ranges.

Generated automatically by AQbD Studio V1. Aligned with ICH Q8/Q9/Q14 Enhanced Lifecycle Guidelines.